

Ben-Gurion University of the Negev
Electrical and Computer Engineering

Satellite and Mobile Communication

Final Project

Submitted by:

Shahar Lavi

Itamar Guerchon

Contents

1	Performance Comparison - MATLAB Model vs. Python Implementation	2
1.1	Signal Spectrum and Spectral Regrowth	2
1.2	HPA Characteristics (AM/AM & AM/PM)	3
1.3	Constellation Before and After Amplification	3
1.4	Non-Linear Distortions and AM/PM Dispersion	4
1.5	Comparison Summary	5
2	Satellite-Ground Laser Link Budget Analysis - MATLAB vs. Python Implementation	6
2.1	Rx Power & Slant Range vs. Elevation	6
2.2	Link Margin Analysis	7
2.3	Loss Breakdown	7
2.4	Comparison Summary	7
3	Evaluation of Multi-Satellite Cooperative Coverage for LEO Constellations	8
3.1	Introduction	8
3.2	System Performance Analysis	8
3.3	Data Representation	10
3.4	Conclusion	11
4	Computational Exercises	12
4.1	Question 1: Downlink Shannon Capacity under Interference	12
4.2	Question 2: ITU Equivalent Power Flux Density (EPFD) Compliance	12
4.3	Question 3: Phased Array Antenna Peak Gain	13

1 Performance Comparison - MATLAB Model vs. Python Implementation

To validate the implementation developed in Python, we compared the results against the standard MATLAB model (`commrfsatlink`). The objective was to ensure that our code, which explicitly implements the physical equations instead of relying on predefined MATLAB blocks, accurately replicates the behavior of the RF chain.

The simulation was run under system parameters representing a typical satellite channel:

- **Modulation and Pulse Shaping:** Data transmission using QAM-16 with Root Raised Cosine (RRC) filtering, roll-off factor $\beta = 0.25$.
- **Link Budget:** Carrier frequency of 4 GHz (C-band), satellite altitude of 35,600 km (geostationary orbit), and antenna diameters of 0.4 m.
- **High Power Amplifier (HPA):** Analytical model of a TWTA amplifier based on Saleh's model. The operating point is controlled via the Back-off parameter.
- **Noise Model:** Addition of AWGN noise calculated according to Boltzmann's equation and a system temperature of 20 K (later adjusted for comparison with MATLAB).

1.1 Signal Spectrum and Spectral Regrowth

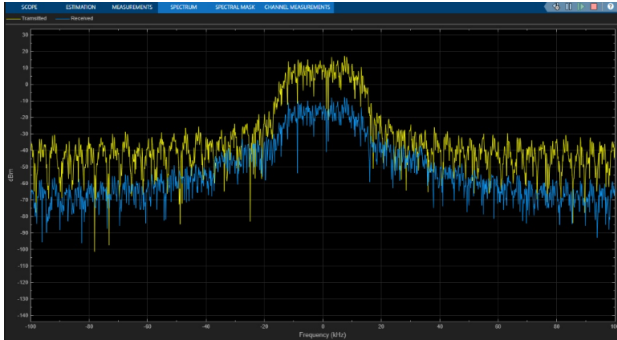


Figure 1: Spectrum in MATLAB



Figure 2: Spectrum in Python

Simulation Parameters (Python):

- `HPA_BACKOFF_DB = 7 dB` (Operating in saturation to exhibit distortions).
- `ENABLE_DPD = False` (No correction algorithm).

Results Analysis: The graphs display the Power Spectral Density (PSD) before the amplifier (yellow) and after it (blue). It can be observed that the RRC filtering cuts off the signal as expected (yellow graph). In the blue graph, the Spectral Regrowth phenomenon is visible in both software environments - the saturated amplifier causes spectral expansion and power leakage into adjacent frequencies, raising the out-of-band noise floor.

Technical Note on Axes: There is a difference in the Y-axis values between the graphs due to normalization differences. MATLAB displays absolute power in dBm, while Python displays normalized density in dB/Hz. Additionally, the visual noise floor is affected by the Resolution Bandwidth (RBW) in each software. Despite the differences in values, the ratio between the central signal peak and the height of the "shoulders" on the sides remains similar in both environments.

1.2 HPA Characteristics (AM/AM & AM/PM)

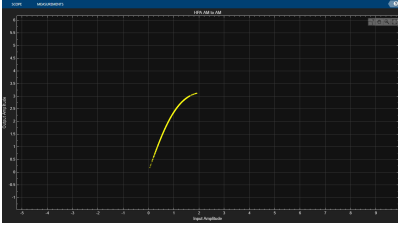


Figure 3: AM/AM MATLAB

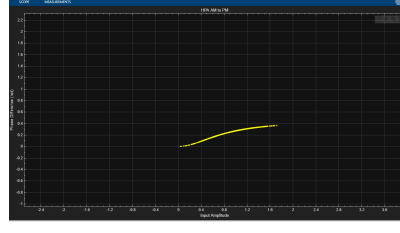


Figure 4: AM/PM MATLAB

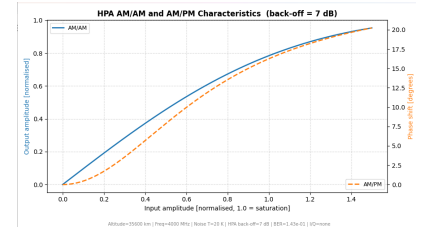


Figure 5: Saleh Curve in Python

Results Analysis: The curves present the amplitude compression (AM/AM) and phase rotation (AM/PM) according to the Saleh model. The graphs show that at low amplitudes the amplifier behaves linearly, followed by significant saturation.

Technical Note on Axes: The X-axis in MATLAB shows raw voltage values (up to around 7 dB), whereas in Python we chose to normalize the input amplitude to an average power of 1. Despite the scale differences, the saturation onset point and the compression ratios derived from the coefficients behave similarly in both environments.

1.3 Constellation Before and After Amplification

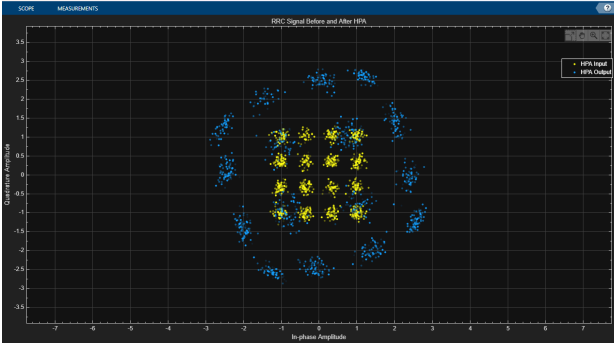


Figure 6: MATLAB: RRC Signal Before & After HPA (Back-off = 7 dB)

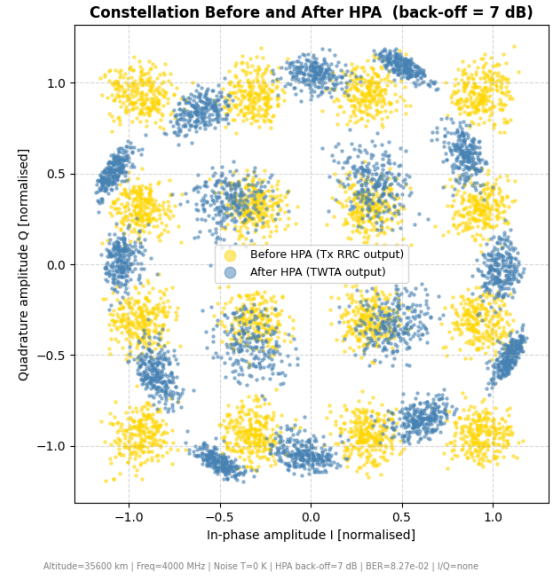


Figure 7: Python: RRC Signal Before & After HPA (Back-off = 7 dB)

Simulation Parameters (Python):

- `HPA_BACKOFF_DB = 7` dB (Operating in saturation).
- `NOISE_TEMPERATURE_K = 0` (No thermal noise, to isolate net distortion).

Results Analysis: To examine the geometric distortion, we extracted a noise-free constellation (before the amplifier in yellow, after it in blue). In QAM-16 modulation, the four corner symbols require Peak Power. The inner symbols remain nearly unchanged, while the corner symbols are compressed inward

and rotated due to TWTA saturation.

Technical Note on Normalization: The constellation display in MATLAB operates on an integer grid (amplitude exceeds 3). In Python, the signal is normalized to a power of 1 (division by $\sqrt{10}$), hence the ideal peak is around 0.948. Nevertheless, the compression percentage of the corners relative to their original position remains similar.

1.4 Non-Linear Distortions and AM/PM Dispersion

To fully characterize the TWTA model, we evaluated the system under a severe non-linear operating point (1 dB back-off), isolating the deterministic non-linear effects by setting the thermal noise temperature to 0 K.

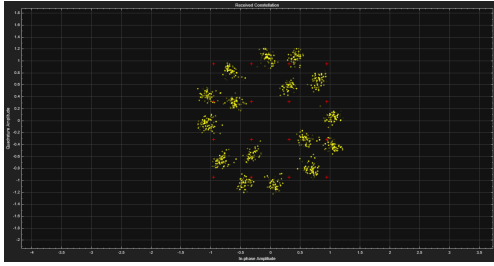


Figure 8: MATLAB: End-to-End (Back-off = 1 dB)

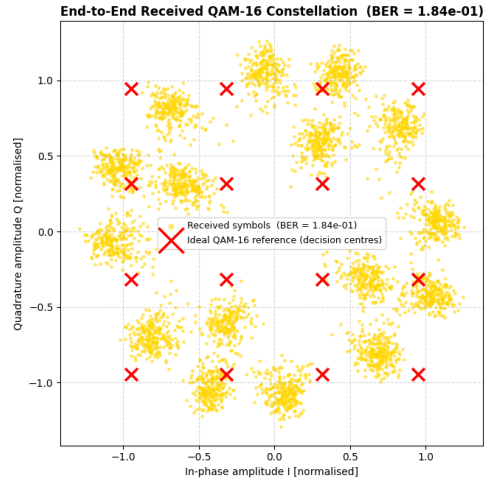


Figure 9: Python: End-to-End (Back-off = 1 dB)

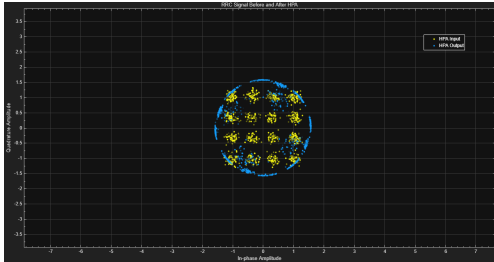


Figure 10: MATLAB: RRC Signal Before & After HPA (Back-off = 1 dB)

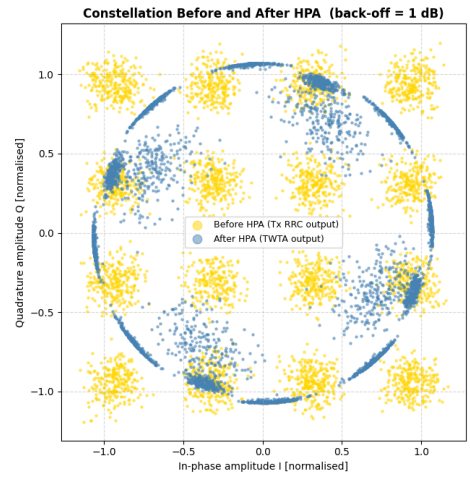


Figure 11: Python: RRC Signal Before & After HPA (Back-off = 1 dB)

Results Analysis:

- **End-to-End Performance (Figures 8 & 9):** Operating at a 1 dB back-off forces the TWTA into severe non-linearity. The outer QAM-16 symbols (which carry the highest peak power) are compressed inwards (AM/AM) and rotated (AM/PM) relative to their ideal decision centers (red crosses). This rigid geometric displacement drives the symbols across decision boundaries, generating an error rate ($\text{BER} \approx 1.84 \cdot 10^{-1}$) independent of thermal noise.

- **Internal AM/PM Dispersion (Figures 10 & 11):** To isolate the mechanism driving the end-to-end errors, the second row overlays the signal immediately before the TWTA (yellow) and after it (blue). The Root Raised Cosine (RRC) filter introduces continuous amplitude fluctuations (PAPR) into the signal envelope. As these varying amplitudes pass through the Saleh model, they are translated into varying phase shifts. As a result, the outer symbols smear into distinct circular arcs.

Technical Note on Normalization: A minor difference in the axis scales can be observed between the MATLAB and Python dispersion plots. This occurs because the Python implementation normalizes the baseband QAM-16 constellation to an average power of 1, whereas MATLAB uses its default unnormalized grid. Despite this visual scaling difference, the relative AM/AM compression depth and the AM/PM rotation angles remain identical in both environments.

1.5 Comparison Summary

The comparison shows that our Python implementation successfully reproduces the behavior of the MATLAB communication model. The numerical discrepancies appearing in the graphs (axis scales and noise powers) are the result of different normalization methods rather than bugs in the code.

2 Satellite-Ground Laser Link Budget Analysis - MATLAB vs. Python Implementation

To validate the Python application developed for the satellite laser communication link, we compared its results against the reference SGLC-Calc MATLAB tool. The objective was to ensure that our Python code accurately implements the physical models for free-space propagation loss (FSPL), optical antenna efficiency, pointing loss, and atmospheric attenuation under various weather conditions and turbulence.

The simulation was executed using the following system parameters, representing a typical Low Earth Orbit (LEO) downlink scenario under clear sky conditions:

- **Global Parameters:** Wavelength of 1550 nm, Transmit Power of 20 dBm, Link Direction set to Downlink (Sat-to-Ground).
- **Ground Station:** Altitude of 0.1 km, Antenna Aperture of 0.8 m, Optical Efficiency of -0.97 dB, and Pointing Error of 1e-06 rad.
- **Satellite:** Orbit Altitude of 500 km, Antenna Aperture of 0.08 m, Optical Efficiency of -0.97 dB, and Pointing Error of 1e-06 rad.
- **Channel & Environment:** Clear Sky condition, with a Required Signal Power (Receiver Sensitivity) threshold of -36.0 dBm.

2.1 Rx Power & Slant Range vs. Elevation

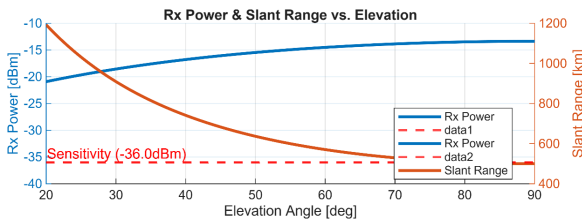


Figure 12: MATLAB: Rx Power vs Elevation

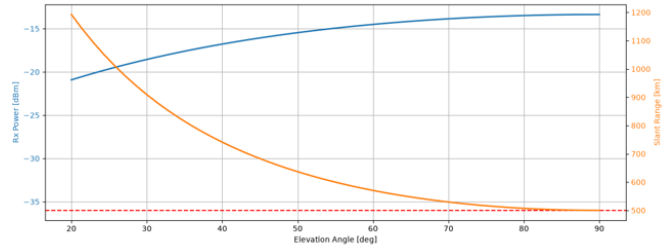


Figure 13: Python: Rx Power vs Elevation

This graph presents the received signal power, Rx Power on the left Y-axis, and the physical distance between the satellite and the ground station, Slant Range on the right Y-axis, as a function of the satellite's elevation angle. As shown, there is an inverse relationship between these variables: as the elevation angle approaches the zenith (90 degrees), the physical distance shortens and atmospheric attenuation decreases, leading to a significant increase in the received signal power. The dashed red line represents the required receiver sensitivity, the intersection point between the Rx Power curve and the sensitivity line defines the minimum elevation angle required to maintain reliable communication. A comparison with the original MATLAB model shows complete agreement in the trends, with minor differences stemming solely from the graphical rendering properties of the Matplotlib library compared to MATLAB's App Designer.

2.2 Link Margin Analysis

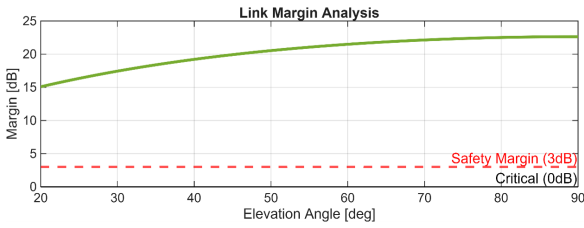


Figure 14: MATLAB: Link Margin Analysis

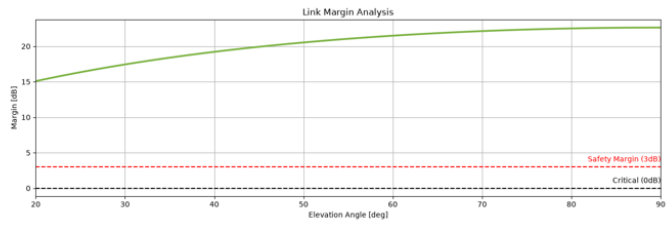


Figure 15: Python: Link Margin Analysis

The Link Margin analysis graph evaluates the stability of the communication channel above the receiver's sensitivity threshold. The calculated value is defined as the difference (in dB) between the actual received power and the receiver sensitivity. Two reference thresholds are defined in the graph: a critical threshold of 0 dB (dashed black line), below which the link is infeasible, and a safety margin of 3 dB (dashed red line), which ensures stable communication that is resilient to minor atmospheric fluctuations. The Python simulation results perfectly match the MATLAB engine, demonstrating how the link margin significantly improves at higher elevation angles under the specified weather conditions.

2.3 Loss Breakdown

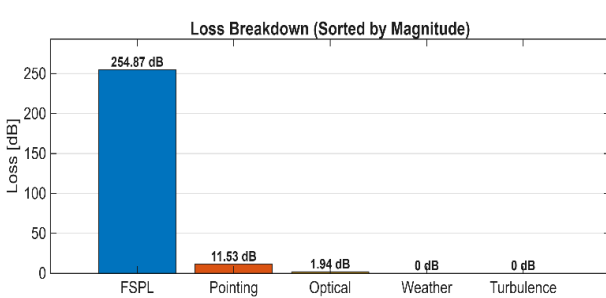


Figure 16: MATLAB: Loss Breakdown

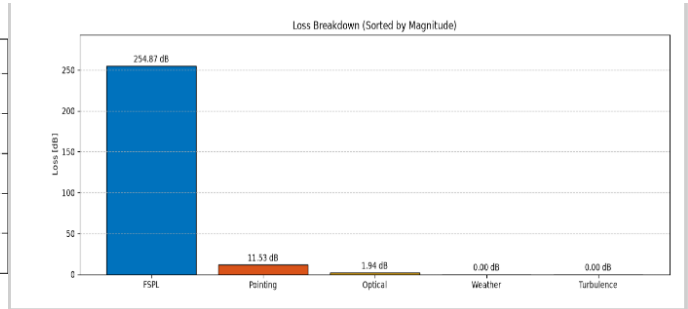


Figure 17: Python: Loss Breakdown

This graph presents a Pareto analysis of the various attenuation factors in the system (Loss Breakdown), sorted in descending order of magnitude. This analysis is critical for identifying bottlenecks in the link budget. As expected from the theoretical model, the most dominant attenuation is the Free Space Path Loss (FSPL), derived directly from the wavelength and the vast slant range. This is followed by losses due to pointing errors, optical efficiency, and atmospheric attenuation (weather and turbulence), depending on the selected scenario. The output generated by the Python code displays a loss distribution completely identical to the MATLAB output, validating the accurate implementation of the physical equations and the successful conversion of the system.

2.4 Comparison Summary

The comparative analysis confirms that our Python implementation successfully reproduces the complex link budget calculations of the baseline MATLAB SGLC-Calc engine. The numerical discrepancies observed in the visualizations are entirely cosmetic, resulting from the differing default normalization grids, padding, and auto-scaling behaviors of Python's Matplotlib compared to MATLAB's plotting engine, rather than any algorithmic deviations in the physical layer implementation.

3 Evaluation of Multi-Satellite Cooperative Coverage for LEO Constellations

3.1 Introduction

The deployment of Low Earth Orbit (LEO) satellite mega-constellations introduces severe spectrum-sharing challenges. Specifically, systems utilizing full-frequency multiplexing (where all beams share the entire available frequency band to improve spectrum utilization) suffer from severe co-layer interference ($\mathcal{I}_{con}$) among adjacent LEO beams, as well as cross-layer interference ($\mathcal{I}_{gso}$) when coexisting with Geostationary Orbit (GSO) networks.

This section evaluates the Cooperative Coverage Scheme (CCS) based on the paper "*Resource Optimization for LEO Constellation Networks: A Multi-Satellite Cooperative Coverage Design*", authored by Xu Ma, Haijun Zhang, Wei Song, Yang Lu, Yuan Wu, and Victor C. M. Leung (2025). To address the interference problem, the authors propose an Improved P-center algorithm to optimize beam placement, replacing traditional approaches like K-means clustering. While the original paper focuses heavily on network-layer resource allocation, our project's objective is to demonstrate its direct impact on the physical RF layer. By simulating the end-to-end RF link, we evaluate how the reduction in co-layer interference translates into measurable improvements in BER and theoretical network throughput.

3.2 System Performance Analysis

To provide a fair comparison under identical channel conditions, we simulated the downlink transmission for both the baseline state-of-the-art method (K-means beam placement with Max-SNR association) and the proposed CCS method. Based on the paper's interference model, the baseline approach suffers from significant beam overlap, which we modeled as a high co-layer interference level of -132 dBW. The proposed CCS minimizes this overlap, reducing the interference from -132 dBW to -150 dBW. These specific values reflect the physical antenna gain patterns: the baseline's overlap exposes users to the high-gain main lobe of interfering beams, whereas the CCS algorithm spaces out beam centers, pushing interference into lower-gain side lobes. Cross-layer GSO interference and thermal noise were kept constant for both scenarios.

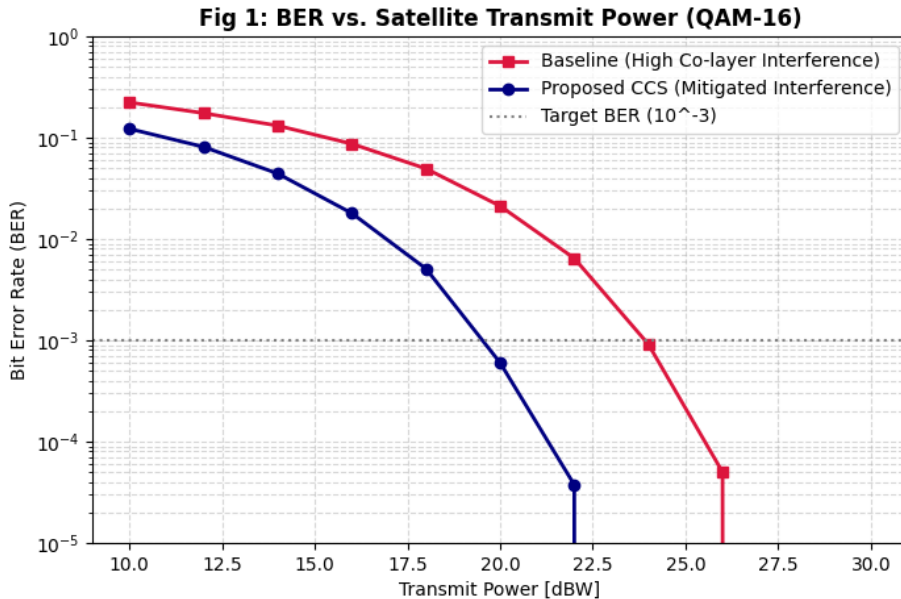


Figure 18: BER vs. Transmit Power demonstrating the mitigation of Co-layer Interference

The graph in Figure 18 illustrates the noise performance and signal conversion efficiency of both systems. The baseline configuration represents an interference-limited channel; the high interference floor flattens the BER curve, requiring high transmit power (approximately 28 dBW) to eliminate errors. In contrast, the CCS effectively mitigates interference, reducing interference such that thermal noise becomes the dominant impairment. For a target BER of 10^{-3} , the CCS requires roughly 4 dB less power than the baseline. By achieving lower BER at lower transmit powers, the CCS enables more efficient resource utilization and supports low-latency communication services in future satellite networks.

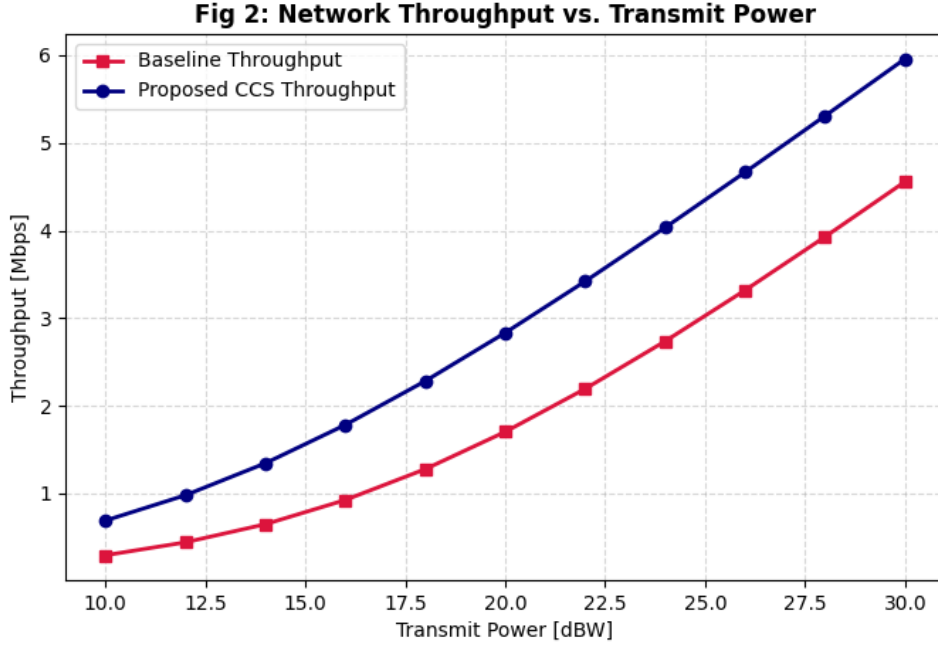


Figure 19: Network Throughput comparison as a function of Transmit Power

This improvement in the Signal-to-Interference-plus-Noise Ratio (SINR) directly affects the system's bandwidth efficiency. Using the Shannon-Hartley theorem, Figure 19 translates the SINR into achievable network throughput. The proposed CCS consistently provides higher data rates for the same transmit power. This aligns with the paper's primary objective: maximizing the network's throughput utility (a metric reflecting the effective data rate fulfilling user demands). By optimizing beam placement to reduce interference, the available satellite power is converted into useful throughput much more efficiently. As a result, a larger fraction of the transmitted power contributes to data delivery.

3.3 Data Representation

To visualize the physical impact of the interference mitigation, we analyzed the raw received symbols before the demodulation decision stage.

Fig 3: Visual SINR Comparison at 20 dBW Transmit Power

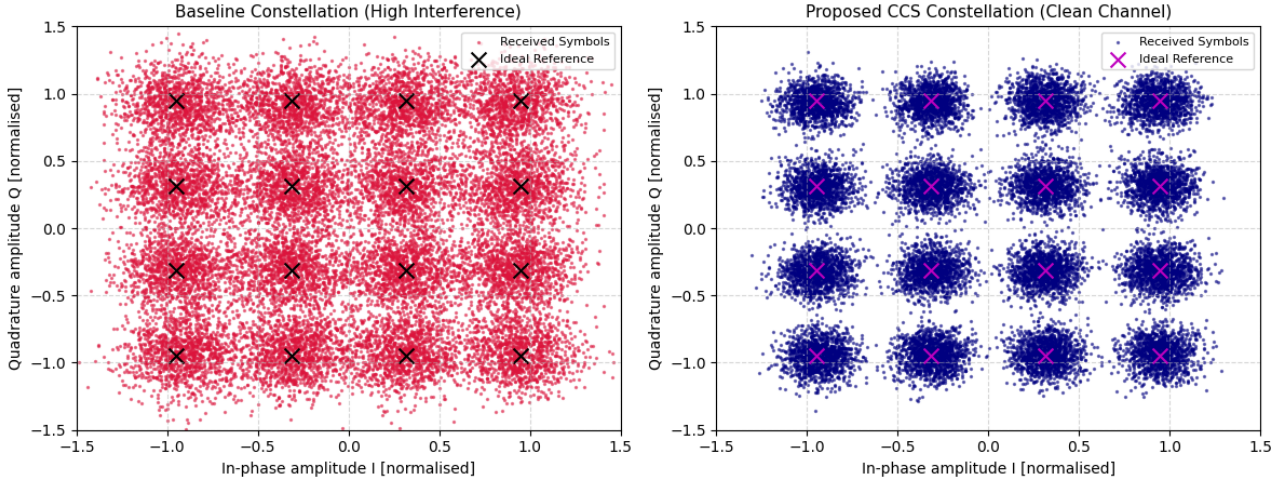


Figure 20: Visual SINR comparison showing the effect of interference on the QAM-16 constellation at a 20 dBW transmit power

Figure 20 presents the QAM-16 received constellation at a specific transmit power of 20 dBW. In the baseline scenario (left), the severe co-layer interference acts as an additional noise source, significantly increasing the variance of each symbol cloud. Consequently, the symbols spread outward and frequently cross the geometric decision boundaries. Conversely, the CCS constellation (right) displays tight, well-defined clusters. By algorithmically separating the overlapping beams, the interference variance is minimized, preventing symbol blurring and reducing symbol dispersion in the received signal.

Fig 4: GSO Interference Protection (EPFD vs Time)

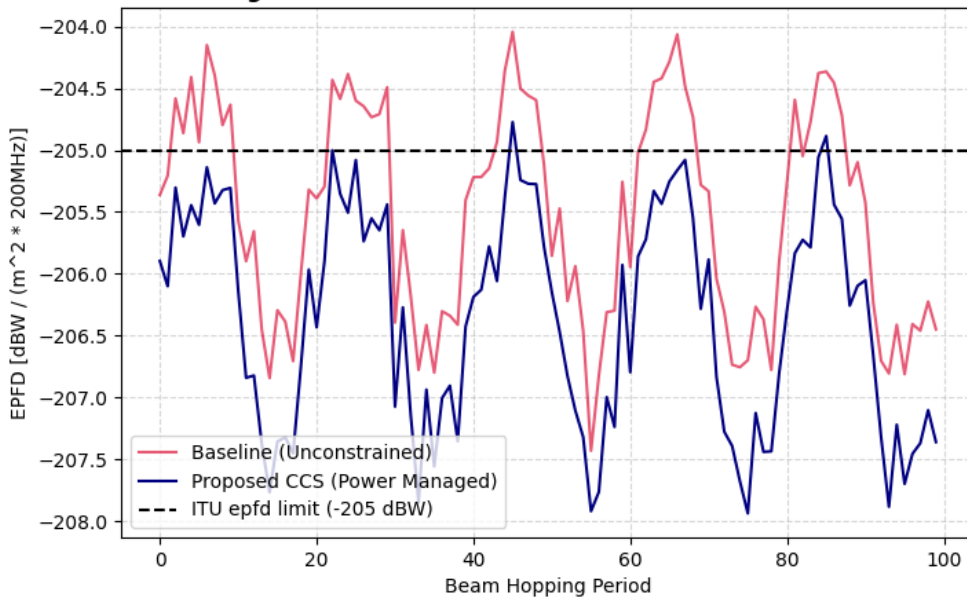


Figure 21: GSO Interference Protection demonstrating CCS compliance with the ITU -205 dBW limit

Beyond internal network efficiency, the paper addresses the critical regulatory constraint of protecting

existing GSO satellites. According to International Telecommunication Union (ITU) regulations, the Equivalent Power Flux Density (EPFD) from a LEO satellite must not exceed -205 dBW at the GSO earth station (denoted as constraint C7 in the paper).

Figure 21 simulates this dynamic scenario over multiple beam-hopping time slots. The baseline approach, which lacks cross-layer awareness, periodically violates this threshold as the LEO satellite moves along its orbit relative to the earth station. The CCS, however, employs a joint beam association and power allocation scheme that actively suppresses the transmit power when approaching the GSO line-of-sight. The CCS curve remains below the ITU threshold throughout the simulated time slots, indicating compliance with the regulatory constraint.

Table 1 outlines the architectural differences and system trade-offs between the two approaches:

Table 1: Architectural Comparison and System Trade-offs: Baseline vs. CCS

System Feature / Metric	Baseline Architecture	Proposed CCS Architecture
Beam Placement	K-means clustering (Prone to high overlap)	Improved P-center algorithm (Minimizes boundary overlap)
GSO Protection	Blind transmission (High risk of ITU violation)	Active power suppression (Strict EPFD compliance)
Resource Allocation	Static / Proportional	Dynamic (Joint beam, power, and time optimization)
Computational Complexity	Low (Standard clustering heuristic)	Moderate (Iterative Block Coordinate Descent)
Scalability (High Traffic)	Severe degradation due to unmanaged interference	Robust performance via proactive interference mitigation

As Table 1 shows, the improved performance of the CCS comes with higher algorithmic complexity. Given the strict ITU constraints, this computational cost is unavoidable.

3.4 Conclusion

In this project, we investigated how network-level resource allocation affects physical RF link performance. Using our custom Python RF simulator, we evaluated the Cooperative Coverage Scheme (CCS) proposed by Ma et al. (2025).

The simulation results confirm that optimizing beam placement and power allocation directly impacts LEO satellite communication. The CCS approach effectively mitigated co-layer interference, improving the Signal-to-Interference-plus-Noise Ratio (SINR). This resulted in a lower Bit Error Rate (BER) and higher achievable throughput compared to the baseline methods. Moreover, the dynamic power management ensured safe coexistence with legacy GSO systems by adhering to ITU EPFD constraints. These results suggest that similar interference management techniques could be useful in future satellite systems, particularly in 6G and IoT scenarios. Future work could extend this physical-layer simulation to tri-layer multi-orbit architectures (LEO, MEO, and GSO), or integrate machine learning models to predict traffic fluctuations and proactively manage interference under dynamic channel conditions.

4 Computational Exercises

4.1 Question 1: Downlink Shannon Capacity under Interference

Problem: A LEO satellite transmits a signal to a user with an effective received signal power of $P_{sig} = 1.5 \cdot 10^{-12}$ W. The system operates with a bandwidth of $B = 200$ MHz. The user experiences co-layer interference $\mathcal{I}_{con} = 4.0 \cdot 10^{-13}$ W from adjacent LEO beams and cross-layer interference $\mathcal{I}_{gso} = 1.0 \cdot 10^{-13}$ W from a GSO satellite. Assuming thermal noise power (KTB) is $5.0 \cdot 10^{-13}$ W, calculate the achievable downlink capacity in Mbps.

Solution: To determine the achievable data rate, we use the Shannon-Hartley capacity formula tailored for the coexistence scenario outlined in the paper (Equation 8). The total interference and noise power in the denominator severely limits the capacity:

$$C = B \log_2 \left(1 + \frac{P_{sig}}{\mathcal{I}_{con} + \mathcal{I}_{gso} + KTB} \right)$$

Substituting the given values:

$$\begin{aligned} C &= 200 \cdot 10^6 \log_2 \left(1 + \frac{1.5 \cdot 10^{-12}}{4.0 \cdot 10^{-13} + 1.0 \cdot 10^{-13} + 5.0 \cdot 10^{-13}} \right) \\ C &= 200 \cdot 10^6 \log_2 \left(1 + \frac{1.5 \cdot 10^{-12}}{10.0 \cdot 10^{-13}} \right) = 200 \cdot 10^6 \log_2(1 + 1.5) \\ C &\approx 200 \cdot 10^6 \cdot 1.3219 \approx 264.38 \text{ Mbps} \end{aligned}$$

Conclusion: Reducing co-layer interference increases the achievable channel capacity.

4.2 Question 2: ITU Equivalent Power Flux Density (EPFD) Compliance

Problem: To protect GSO Earth Stations, ITU regulations mandate that the EPFD must not exceed -205 dBW. A LEO satellite transmits with a power $p_b = 15$ W. Its channel gain towards the GSO station is $g = 1$, and the distance is $d = 1000$ km. If the GSO station's maximum receiver gain is $G_{max} = 10^4$ (linear), calculate the resulting EPFD. Does this assignment violate the ITU constraint?

Solution: The EPFD evaluates the interference severity at the GSO receiver. Using the physical link model provided in the paper (Equation 19):

$$epfd = 10 \log_{10} \left(\frac{p_b \cdot g}{4\pi d^2 G_{max}} \right)$$

Converting the distance to meters ($d = 10^6$ m) and substituting the values:

$$\begin{aligned} epfd &= 10 \log_{10} \left(\frac{15 \cdot 1}{4\pi \cdot (10^6)^2 \cdot 10^4} \right) = 10 \log_{10} \left(\frac{15}{4\pi \cdot 10^{16}} \right) \\ epfd &= 10 \log_{10}(1.1936 \cdot 10^{-16}) \approx -159.23 \text{ dBW} \end{aligned}$$

Conclusion: The calculated EPFD is -159.23 dBW. Since -159.23 dBW is much greater (stronger interference) than the strict ITU limit of -205 dBW, this static beam assignment **violates** the constraint. This result demonstrates that static power allocation may fail to satisfy the ITU EPFD constraint.

4.3 Question 3: Phased Array Antenna Peak Gain

Problem: The LEO satellite payload utilizes a multi-beam phased array antenna. Given an antenna consisting of $N = 400$ cells, an efficiency of $\eta = 0.6$, and a half-power beamwidth (HPBW) of $\theta_{3dB} = 3^\circ$, calculate the theoretical peak antenna gain G_{max} in dBi.

Solution: According to the antenna model (Equation 3), the peak gain is inversely proportional to the square of the HPBW in radians. First, we convert θ_{3dB} from degrees to radians:

$$\theta_{rad} = 3 \cdot \frac{\pi}{180} \approx 0.05236 \text{ rad}$$

Next, we calculate the peak linear gain:

$$G_{max} = \frac{\eta N^2 \pi^2}{(\theta_{rad})^2} = \frac{0.6 \cdot (400)^2 \cdot \pi^2}{(0.05236)^2} = \frac{947,482.0225}{0.002741} \approx 3.456 \cdot 10^8 \text{ (linear)}$$

Finally, we convert the linear gain to decibels isotropic (dBi):

$$G_{dBi} = 10 \log_{10}(3.456 \cdot 10^8) = 10 \cdot 8.538 \approx 85.38 \text{ dBi}$$

Conclusion: The calculated peak gain is approximately 85.38 dBi, which reflects the high directivity of phased-array antennas used in modern LEO satellites.